

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
Work Integrated Learning Programmes Division
First Semester 2025-2026
Comprehensive examination
(EC-3 MakeUp) -Answer Key

Course No. : AIMLCZC418
Course Title : Introduction to Statistical Methods
Nature of Exam : Open Book
Weightage : 40%
Duration : 2 Hours
Date of Exam : 07-03-2026_FN

No. of Pages	= 3
No. of Questions	= 8

Note to Students:

1. Please follow all the Instructions to Candidates given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Q1. a) An organization evaluates two training programs for improving employee productivity.
Program A: 20 employees, mean productivity score = 81, standard deviation = 9
Program B: 25 employees, mean productivity score = 77, standard deviation = 8. At the 5% level of significance, test whether Program A leads to significantly higher productivity.
[3 marks]

b) A company produces light bulbs with an average lifespan of 1000 hrs. and a standard deviation of 50 hrs. To estimate the population mean lifespan, a random sample of 40 bulbs is selected and their average lifespan is found to be 990hrs. Calculate a 99% Confidence interval for the population mean lifespan of the light bulbs. **[2 Marks]**

Solution:

a).

Given Data

Program A:

$$n_1 = 20, \bar{X}_1 = 81, s_1 = 9$$

Program B:

$$n_2 = 25, \bar{X}_2 = 77, s_2 = 8$$

Significance level: $\alpha = 0.05$

Null Hypothesis (H_0): $\mu_1 \leq \mu_2$

(Program A is not better than Program B)

Alternative Hypothesis (H_1): $\mu_1 > \mu_2$

(Program A gives higher productivity)

This is a **right-tailed test**.

Test Statistic:

$$t_{cal} = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = 1.56$$

$$df = n_1 + n_2 - 2 = 20 + 25 - 2 = 43$$

At 5% level (one-tailed) $t_{tab} = 1.68$

Conclusion: $t_{cal} < t_{tab}$ At the **5% significance level**, we **fail to reject** H_0 .

At the 5% significance level, there is not enough evidence to conclude that Program A leads to significantly higher productivity than Program B.

1.b.

Given $\bar{x} = 990, \sigma = 50$ and $n = 40$.

we use the Z-confidence interval formula $\bar{x} \pm Z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

For 99% confidence level $Z_{\alpha/2} = 2.576$

Confidence Interval is (969.62, 1010.38).

The **99% confidence interval for the population mean lifespan** of the light bulbs is

969.62 hours to 1010.38 hours.

Q2. A company wants to know whether **device type** influences **purchase decision**.

Device	Purchased	Not Purchased	Total
Mobile	80	120	200
Laptop	70	50	120
Tablet	50	30	80

Use a chi-square test to determine whether device type and purchase decision are independent ($\alpha = 0.05$). **[5 marks]**

Solution:

H₀: Device type and purchase decision are independent.

H₁: Device type and purchase decision are not independent.

LOS: $\alpha = 0.05$

Device	Purchased	Not purchased	Total
Mobile	80	120	200
Laptop	70	50	120
Tablet	50	30	80

Device	Purchased	Not purchased	Total
Total	200	200	400

Expected Frequencies $E = \frac{(\text{Row Total})(\text{Column Total})}{\text{Grand Total}}$

Mobile Purchased	100
Mobile Not Purchased	100
Laptop Purchased	6
Laptop Not Purchased	60
Tablet Purchased	40
Tablet Not Purchased	40

O's	E's	$\frac{(O - E)^2}{E}$
80	100	4
120	100	4
70	60	1.67
50	60	1.67
50	40	2.5
30	40	2.5
	Total	= 16.34

$$\chi^2 = \sum \frac{(O-E)^2}{E} = 16.34$$

$$\text{dof} = (r-1)(c-1) = 2, \quad \chi^2_{calc} = 5.991$$

χ^2 calculated = 16.34 > χ^2 tab = 5.991 at $\alpha = 0.05$ for 2 dof

Decision : we reject the Null hypothesis H_0 at $\alpha = 5\%$ LOS.

i.e, we accept H_1 .

3Q. The annual production (in thousand units) of a factory from 2017 to 2024 is given below:
[5 marks]

Year	2017	2018	2019	2020	2021	2022	2023	2024
Production	42	46	39	35	37	44	51	58

Compute a **3-year moving average** and use it to **forecast production for 2025**.

Solution:

The 3-year moving average smooths annual production data to forecast 2025 output at 51 thousand units.

Year	Production (thousand units)	3-Year Moving Average
2017	42	-
2018	46	-
2019	39	42.33
2020	35	40
2021	37	37
2022	44	38.67
2023	51	44
2024	58	51

Forecast for 2025

Using the **last 3-year moving average (2022–2024)**:

. Forecast Production for 2025 = 51 thousand units

4Q. An IoT system records the daily average temperature (°C) of a server room over 10 days:

Day	1	2	3	4	5	6	7	8	9	10
Temp	22.1	22.4	22.6	22.9	23.1	23.4	23.7	23.9	24.1	24.3

- Compute the lag-1 autocorrelation coefficient.
- Comment on whether the temperature series exhibits persistence.

Solution:

- Compute the lag-1 autocorrelation coefficient

Formula:

$$r_1 = \frac{\sum_{t=2}^n (X_t - \bar{X})(X_{t-1} - \bar{X})}{\sum_{t=1}^n (X_t - \bar{X})^2}$$

$$\bar{X} = 23.25$$

$$\sum (x_i - \bar{x})(x_{i-1} - \bar{x}) = 3.5925 \quad \text{and} \quad \sum (x_i - \bar{x})^2 = 5.085$$

$$\text{Then } r_1 = \frac{\sum (x_i - \bar{x})(x_{i-1} - \bar{x})}{\sum (x_i - \bar{x})^2} = 0.71$$

(b) Comment on Persistence

Since the **lag-1 autocorrelation coefficient is positive and relatively high (0.71)**, the temperature series shows **strong positive persistence**.

- This indicates a **strong positive correlation** between the temperatures of consecutive days.
- Therefore, the temperature series **exhibits significant persistence**, meaning that if the temperature is high on one day, it is highly likely to remain high on the following day, reflecting a steady upward trend in the server room temperature.

Q5. A manufacturing company states that the average weight of packets filled by an automated machine is 500 grams. A random sample of 81 packets drawn from a large production lot has a mean weight of 496 grams. From past quality-control records, the population standard deviation of packet weights is known to be 18 grams. At the 5% level of significance, test whether the machine is underfilling the packets.

[5 Marks]

Solution:

Here, $n = 81$, $\bar{x} = 496$, $\sigma = 18$

$H_0: \mu = 500$ (the machine is not underfilling) versus $H_a: \mu < 500$ (The machine is underfilling)

The level of significance is 0.05

$$\text{The z-statistic is } z = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}} = \frac{496 - 500}{18 / \sqrt{81}} = \frac{-4}{2} = -2$$

$\therefore$ Calculated z value = -2.

we reject null hypothesis and accept H_1 .

The level of significance is 0.05. critical value is ± 1.645

$\therefore$ Calculated z value $|-2|$ i.e., $2 > 1.645$, we reject Null hypothesis

Q6. A small online grocery store tracks the weekly demand (in units) for a popular product.

[5 marks]

The demand shows a steady upward trend, but no seasonal pattern. The observed demand for 7 consecutive weeks is given below:

Week	1	2	3	4	5	6	7
Demand (units)	20	22	24	26	28	30	32

The store wants to forecast future demand using Holt's Double Exponential Smoothing.

Level smoothing constant: $\alpha=0.4$; Trend smoothing constant: $\beta=0.3$;

Initial values: Initial level: $L_1=20$; Initial trend: $T_1=2$

a) Compute the smoothed level (L_t) and smoothed trend (T_t) for weeks 2 to 7.

b) Using the values obtained at week 7, compute the forecast for week 8.

c) Briefly explain why Holt's method is more suitable than simple exponential smoothing for this data.

Solution:

Given:

Initial level, $L_1 = 20$

Initial trend, $T_1 = 2$

Level smoothing constant, $\alpha = 0.4$

Trend smoothing constant, $\beta = 0.3$

We use the following formulas to update the level and trend:

$$L_t = \alpha * \text{Demand}(t) + (1 - \alpha) * (L_{t-1} + T_{t-1})$$

$$T_t = \beta * (L_t - L_{t-1}) + (1 - \beta) * T_{t-1}$$

a) Starting from week 2:

Week 2:

$$L_2 = 0.4 * 22 + (1 - 0.4) * (20 + 2) = 8.8 + 0.6 * 22 = 8.8 + 13.2 = 22$$

$$T_2 = 0.3 * (22 - 20) + (1 - 0.3) * 2 = 0.3 * 2 + 0.7 * 2 = 0.6 + 1.4 = 2$$

Week 3:

$$L_3 = 0.4 * 24 + (1 - 0.4) * (22 + 2) = 9.6 + 0.6 * 24 = 9.6 + 14.4 = 24$$

$$T_3 = 0.3 * (24 - 22) + (1 - 0.3) * 2 = 0.3 * 2 + 0.7 * 2 = 0.6 + 1.4 = 2$$

Week 4:

$$L_4 = 0.4 * 26 + (1 - 0.4) * (24 + 2) = 10.4 + 0.6 * 26 = 10.4 + 15.6 = 26$$

$$T_4 = 0.3 * (26 - 24) + (1 - 0.3) * 2 = 0.3 * 2 + 0.7 * 2 = 0.6 + 1.4 = 2$$

Week 5:

$$L_5 = 0.4 * 28 + (1 - 0.4) * (26 + 2) = 11.2 + 0.6 * 28 = 11.2 + 16.8 = 28$$

$$T_5 = 0.3 * (28 - 26) + (1 - 0.3) * 2 = 0.3 * 2 + 0.7 * 2 = 0.6 + 1.4 = 2$$

Week 6:

$$L_6 = 0.4 * 30 + (1 - 0.4) * (28 + 2) = 12 + 0.6 * 30 = 12 + 18 = 30$$

$$T_6 = 0.3 * (30 - 28) + (1 - 0.3) * 2 = 0.3 * 2 + 0.7 * 2 = 0.6 + 1.4 = 2$$

Week 7:

$$L_7 = 0.4 * 32 + (1 - 0.4) * (30 + 2) = 12.8 + 0.6 * 32 = 12.8 + 19.2 = 32$$

$$T_7 = 0.3 * (32 - 30) + (1 - 0.3) * 2 = 0.3 * 2 + 0.7 * 2 = 0.6 + 1.4 = 2$$

b) The forecast for week 8 is given by:

$$F8 = L7 + T7 = 32 + 2 = 34$$

c) Holt's method is more suitable for this data because it accounts for both the level and the trend in the demand, providing a more accurate forecast when there is a steady upward or downward trend in the data, as is the case here. Simple exponential smoothing, on the other hand, only accounts for the level and does not consider the trend, making it less suitable for data with a significant trend component.

Q7. A dataset measures daily screen time and productivity score:

[5 Marks]

Screen Time (hrs)	1	2	3	4	5	6
Productivity	90	85	78	72	65	60

- Fit a simple regression model
- Interpret direction of relationship
- Predict productivity when screen time = 3.5 hrs.

Solution:

screen time(X)	productivity(y)	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$
1	90	-2.5	15	-37.5	6.25
2	85	-1.5	10	-15	2.25
3	78	-0.5	3	-1.5	0.25
4	72	0.5	-3	-1.5	0.25
5	65	1.5	-10	-15	2.25
6	60	2.5	-15	-37.5	6.25
3.5	75			-108	17.5

a) $\beta_1 = \frac{-108}{17.5} = -6.1714$ (approximately)

$$\beta_0 = 75 - (-6.1714 \times 3.5) = 75 + 21.6 = 96.6$$

the regression equation:

$$y = 96.6 - 6.1714x$$

b) Since $\beta_1 = -6.1714 < 0$, there is a negative relationship between screen time and productivity, meaning that as screen time increases, productivity decreases.

c) **Predict productivity when screen time = 3.5 hrs:**

Using the regression equation $Y = \beta_0 + \beta_1 X$:

$$Y = 96.6 - 6.1714 * 3.5 = 96.6 - 21.6 = 75$$

Q8. A researcher wants to test whether three different teaching methods lead to different average test scores. The following scores are obtained from three groups of students.

Method 1	Method 2	Method 3
15	18	22
17	16	24
16	20	23
14	19	21

At the 5% level of significance, test by using one way ANOVA, whether there is any significant difference in the mean scores of the three teaching methods.

Assume that the populations are normally distributed with equal variances. **[5 Marks]**

Solution:

$H_0: \mu_1 = \mu_2 = \mu_3$

H_1 : At least one mean differs

$\alpha = 0.05$

Group means:

$M1 = 15.5, M2 = 18.25, M3 = 22.5$

Grand Mean = 18.75

ANOVA results:

Source	SS	df	MS	F
Between	99.5	2	49.75	23.88
Within	18.75	9	2.083	
Total	118.25	11		

Critical value: $F_{0.05}(2,9) = 4.26$

Since $F = 23.88 > 4.26$, reject H_0 .

Conclusion: There is a significant difference in the mean scores of the three teaching methods.
